TheLook eCommerce Analytics: Cart Abandonment and Product Returns

Executive Summary

TheLook loses money in two places: shoppers who add items to the cart and then leave (57.94% of cart sessions), and items that come back after delivery (11.79% of items sold). This report follows the customer from the product page to the purchase and then to the return, and checks whether inventory supports the sales.

The five main findings:

- **Sales grow fast.** Gross revenue is $9.22M from 106,617 orders. After returns, net revenue is $8.13M and net profit is $4.22M. Yearly revenue rose from $0.19M in 2019 to $4.37M in 2023.
- **Cart abandonment is the biggest drop, but it improves every year.** Over the whole period 432,146 sessions reached the cart and 181,759 ended in a purchase (42.06%). Abandonment fell from 93% in 2019 to 37% in 2023.
- **The funnel is the same in every channel and browser.** Conversion stays between 26.5% and 27.0%. The problem is the checkout step itself, not one traffic source.
- **Returns are quick and spread evenly.** They cost $570K of profit (11.9% of total profit), happen 1.5 days after delivery on average (never later than 3 days), and stay near 11.8% in every category, department and year.
- **Inventory is about twice the sales.** 308,946 units ($8.85M) are in stock, against 154,669 units sold since 2019. At the forecast demand, the stock covers about 26 months.

The recommended actions are: fix the cart-to-purchase step, improve product information to reduce returns, pause new purchasing until stock falls, and grow repeat purchases. Details are in the Recommendations section.

Business Problem and Objectives

TheLook is an online fashion store. Like most online stores, it has two problems that quietly reduce profit.

Problem 1: Cart abandonment. A shopper finds a product, adds it to the cart, and then leaves without buying. The store paid to bring this shopper in, so every abandoned cart is lost revenue.

Problem 2: Product returns. A customer buys an item and sends it back. The store loses the sale, and also pays for shipping and handling.

The objectives of this analysis are:

1. Measure how many shoppers drop out at each step of the purchase journey (home, department, product, cart, purchase).
2. Measure how often products are returned, how fast, and how much profit this costs.
3. Check whether returns or abandonment are concentrated in a specific category, brand, department, traffic source or year.
4. Check whether inventory is healthy by comparing stock with sales and with forecast demand.
5. Turn these findings into clear actions the business can take.

The analysis is delivered as an Excel workbook with a data model, DAX measures, PivotTables and an interactive dashboard of four pages: Overview, Funnel, Returns and Inventory. This report explains what the dashboard shows.

Data and Method

Data source. The TheLook e-commerce fashion dataset, loaded into Excel as seven tables:

- **Users:** customer details such as country and traffic source.
- **Products:** name, brand, category, department, cost and retail price.
- **Orders:** one row per order, with its dates and status.
- **Order_items:** one row per item in an order, with sale price, status and return date.
- **Inventory_items:** one row per physical unit, with cost and sold date.
- **Events:** the web behavior log (page views, cart, purchase) by session.
- **Distribution_centers:** the warehouses.

Step 1: Data engineering in Power Query. Each table was cleaned and loaded as Connection Only into the Data Model. Missing text values (for example brand or category) were replaced with "Unknown". Missing numbers, dates and keys were left empty, because inventing a value would make the results wrong. No calculated columns were added in Power Query. All calculations are DAX measures.

Step 2: Data model. A Calendar table was created and the tables were joined with One-to-Many relationships. This lets one slicer or timeline filter every page.

Step 3: DAX measures. Revenue, cost, profit, returns, funnel rates, retention and inventory were built as measures (defined in the next section).

Step 4: PivotTables and dashboard. Each chart comes from a PivotTable on the Data Model. Slicers and a timeline are connected across pages so the viewer can filter by year, category, department and traffic source.

Step 5: Forecast. Monthly sales were forecast with Excel Forecast Sheet (ETS, 95% confidence interval, seasonality of 12 months) and compared with current stock.

Sales rules used everywhere in this report:

- Cancelled items are excluded from sales.
- Processing, Shipped and Complete items count as sales.
- Returned items are included in Gross figures and deducted in Net figures.

KPI Definitions

Every KPI below is a DAX measure. Cancelled items are always excluded unless stated.

KPI	What it means	How it is calculated
Total Revenue	Gross sales before returns	Sum of sale price, not Cancelled
Total Cost	What the sold items cost the store	Sum of product cost, not Cancelled
Total Profit	Gross profit	Total Revenue minus Total Cost
Profit Margin %	Profit per dollar of sales	Total Profit divided by Total Revenue
Total Orders	Number of orders	Distinct orders, not Cancelled
AOV	Average order value	Total Revenue divided by Total Orders
Net Revenue	Sales that were kept	Total Revenue minus Returned Revenue
Net Profit	Profit after returns	Total Profit minus profit lost on returned items
Return Rate	Share of items sent back	Returned Items divided by Items Sold
Avg Days To Return	Speed of returns	Average of returned date minus delivered date
Retention Rate	Customers who bought more than once	Repeat Customers divided by Customers
Churn Rate	Customers who bought only once	1 minus Retention Rate
Total Sessions	Number of web visits	Distinct session IDs

KPI	What it means	How it is calculated
Cart Abandonment Rate	Cart sessions with no purchase	(Cart Sessions minus Purchase Sessions) divided by Cart Sessions
Conversion Rate	Sessions that end in a purchase	Purchase Sessions divided by Total Sessions
Cart to Purchase Rate	Carts that become purchases	Purchase Sessions divided by Cart Sessions
Items In Stock	Units not sold yet	Inventory units with no sold date
Stock Value	Money tied up in stock	Sum of cost of unsold units
Stock to Sales Ratio	How many units sit in stock per unit sold	Items In Stock divided by Items Sold
Months of Cover	How long stock lasts	Items In Stock divided by forecast monthly demand

Findings 1: Sales Overview

Dashboard page: Overview

The store is growing and profitable.

Measure	Value
Total Revenue (gross)	$9.22M
Total Orders	106,617
Average order value	about $86
Net Revenue (after returns)	$8.13M
Net Profit (after returns)	$4.22M
Profit margin before returns	about 52%

What this tells us

- Revenue grew from $0.19M in 2019 to $4.37M in 2023. The business is scaling quickly.
- Returns remove about $1.1M of revenue and $0.57M of profit. This is real money, but the business stays strongly profitable.

- Traffic source Search brings about 70% of revenue. This is a strong dependence on one channel.
- We tested whether sales change by day of the week and found no meaningful weekday effect. We also found no support for the idea that one social channel sells more on weekends. We do not recommend weekday-based campaigns from this data.

Findings 2: Web Behavior and Funnel

Dashboard page: Funnel

The funnel follows a visit through five steps: Home, Department, Product, Cart, Purchase. A session counts once at each step it reaches.

Measure	Value
Sessions that reached the Cart	432,146
Sessions that ended in a Purchase	181,759
Cart to Purchase Rate	42.06%
Cart Abandonment Rate	57.94%
Conversion Rate (sessions to purchase)	between 26.5% and 27.0% in every channel

What this tells us

- Over the whole period, more than half of the shoppers who add an item to the cart leave without buying (57.94%). This is the largest loss in the journey. It has improved every year, from 93.11% in 2019 to 37.07% in 2023, so the 2023 level is the right starting point for new targets.
- Conversion is almost identical across traffic sources and browsers. If one channel were weak, we would see it here. We do not, so the cause is most likely the checkout experience that every shopper shares (shipping cost shown late, forced account creation, payment options, and similar).
- This means that spending more on one channel will not fix abandonment. Improving checkout will help every channel at once.

Abandonment by year

Year	Cart Sessions	Purchase Sessions	Cart Abandonment Rate
2019	52,937	3,646	93.11%
2020	62,416	12,692	79.67%

Year	Cart Sessions	Purchase Sessions	Cart Abandonment Rate
2021	74,134	24,306	67.21%
2022	91,915	42,202	54.09%
2023	135,401	85,207	37.07%
2024 (partial)	15,344	13,706	10.68%

Abandonment falls steadily every year, so this looks like a real pattern in the data and not a recording error. 2024 has only the first weeks of January (the data ends on 2024-01-21), so its 10.68% should not be compared with full years.

Data caution. The average session duration is not reliable. In this data one session can span several days (for example from 2024-01-17 to 2024-01-21), so it is not a real visit time and is not used in this report.

Findings 3: Product Returns

Dashboard page: Returns

Measure	Value
Return Rate	11.79% of items sold
Revenue returned	about $1.1M
Profit lost to returns	$570K (11.9% of total profit)
Average days to return	1.5 days after delivery
Longest time to return	3 days

What this tells us

- About 1 item in 9 comes back. The cost is large, but it is predictable.
- Returns happen fast. Almost all are back within three days of delivery, so the store learns about a problem very early.
- The return rate stays near 11.8% in every product category, every department and every year. No category or brand is a clear "problem product". Returns look like a store-wide pattern, not a product-specific one.
- Because the rate is flat, the best levers are store-wide: better size guides, clearer photos and descriptions, and a return policy that discourages ordering several sizes to keep one.

What-if. The Returns page has a Return Reduction % input. As an example, cutting the return rate by one quarter would keep roughly $140K of profit, based on the $570K currently lost. Use the input to test other targets.

Findings 4: Inventory and Forecast

Dashboard page: Inventory

Measure	Value
Units in stock	308,946
Stock value (at cost)	$8.85M
Units sold since 2019	154,669
Stock to Sales Ratio	2.0
Months of Cover (at forecast demand)	about 26 months

What this tells us

- The store holds about two units for every unit it has ever sold. Stock value ($8.85M) is almost as large as total revenue ($9.22M).
- The ratio is about 2.0 in every category and department. Overstock is not limited to a few products. It is a store-wide purchasing pattern.
- The Excel Forecast Sheet (ETS, 95% confidence interval, 12-month seasonality) projects continued monthly growth. Even with this growth, current stock would take about 26 months to sell. A healthy level for fashion, where styles change by season, is much lower.
- Money tied up in stock cannot be used for marketing or checkout improvements, and unsold fashion items lose value over time.

Forecast caution. The forecast comes from a short sales history that grew very fast, so the confidence range is wide. Use it as a direction, not an exact number.

Business Recommendations

1. **Fix the cart-to-purchase step (highest impact).** Show shipping cost and taxes early, offer guest checkout, add more payment options, and send a reminder email to shoppers who leave a full cart. In 2023, 63% of carts ended in a purchase. Target: keep this improving toward 70%. Track it with the Cart Abandonment Rate by year on the Funnel page.

2. **Reduce returns with better product information.** Add size guides, more photos and accurate descriptions. Because returns are even across all categories, start with a store-wide change, then test it on one category first. Track Return Rate and use the What-if input to size the benefit.
3. **Pause new purchasing until stock falls.** With about 26 months of cover, new buying should wait. Clear old stock with promotions and bundles. Track Months of Cover and the Stock to Sales Ratio.
4. **Grow repeat customers.** Retention is the cheapest source of sales. Use post-purchase offers and loyalty rewards. Track Retention Rate and Churn Rate.
5. **Reduce dependence on Search.** About 70% of revenue comes from one source. Test other channels in small budgets, but do not expect them to convert better, because conversion is the same everywhere.

Actions 1 and 3 should come first: one raises sales from traffic the store already has, and the other frees cash.